

Interaksi Manusia dan Komputer (IMK)

Overview
Design

User-Centered

Overview Proses Desain User
Interface



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Overview
Design

User-Centered



Tujuan

Setelah mengikuti materi ini mahasiswa dapat:

1. Mengetahui proses yang dilakukan pada desain User Interface

Overview User Centered Design (UCD)

What is User-Centered Design?

- An approach to UI development and system development.
- Focuses on understanding:
 - Users, and
 - Their goals and tasks, and
 - The environment (physical, organizational, social)
- Pay attention to these throughout development

ISO on User-centered Design

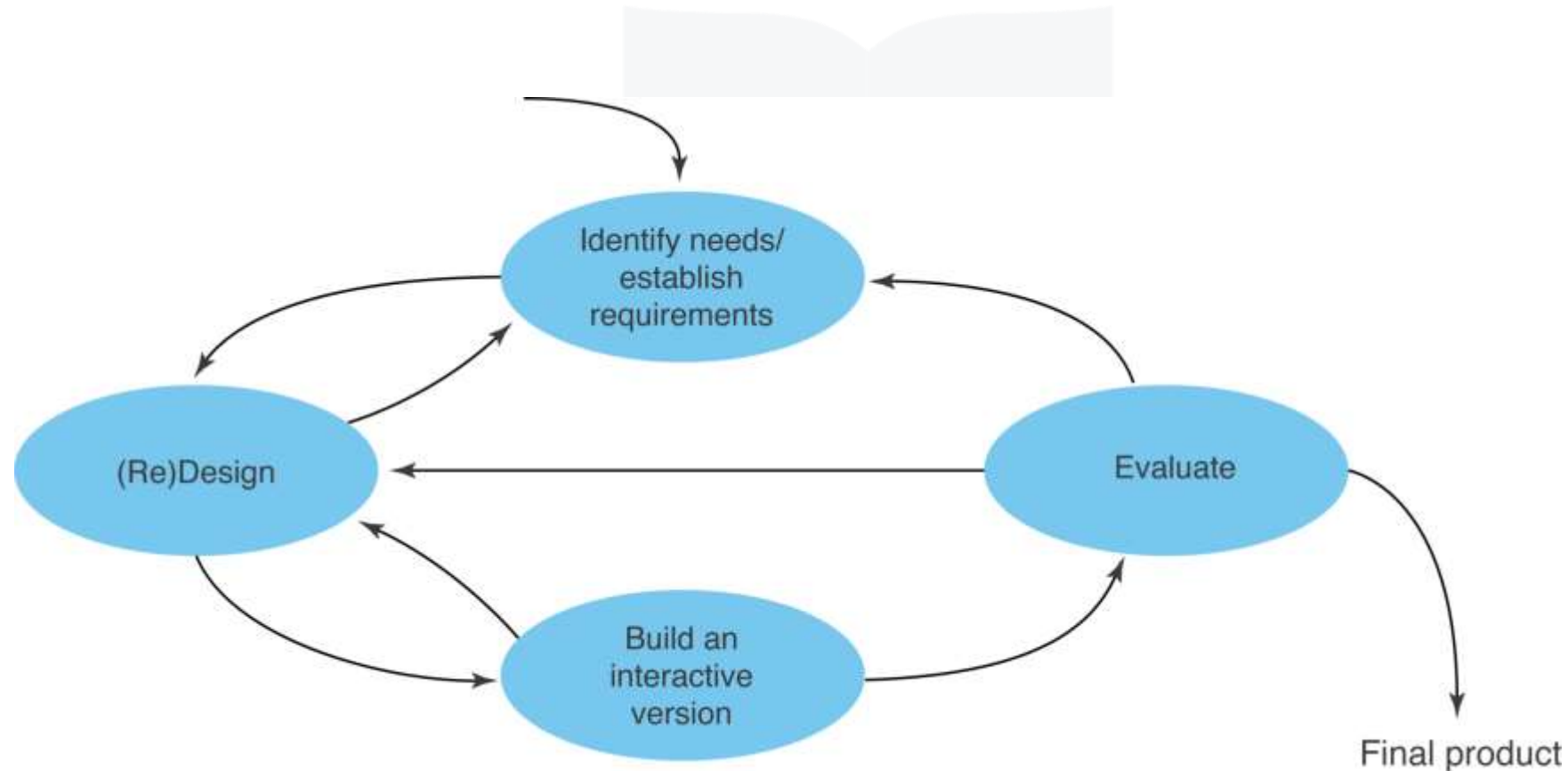
- ISO 13407 describes human-centered design processes for interactive systems
- Principles of human-centered design:
 - Active involvement of users
 - Appropriate allocation of function between user and system
 - Iteration of design solutions
 - Multidisciplinary design teams

What is a user-centered approach?

User-centered approach is based on:

- **Early focus on users and tasks:** directly studying cognitive, behavioral, anthropomorphic & attitudinal characteristics
- **Empirical measurement:** users' reactions and performance to scenarios, manuals, simulations & prototypes are observed, recorded and analysed
- **Iterative design:** when problems are found in user testing, fix them and carry out more tests

A simple interaction design model



Exemplifies a user-centered design approach

1. Identifying needs and establishing requirements

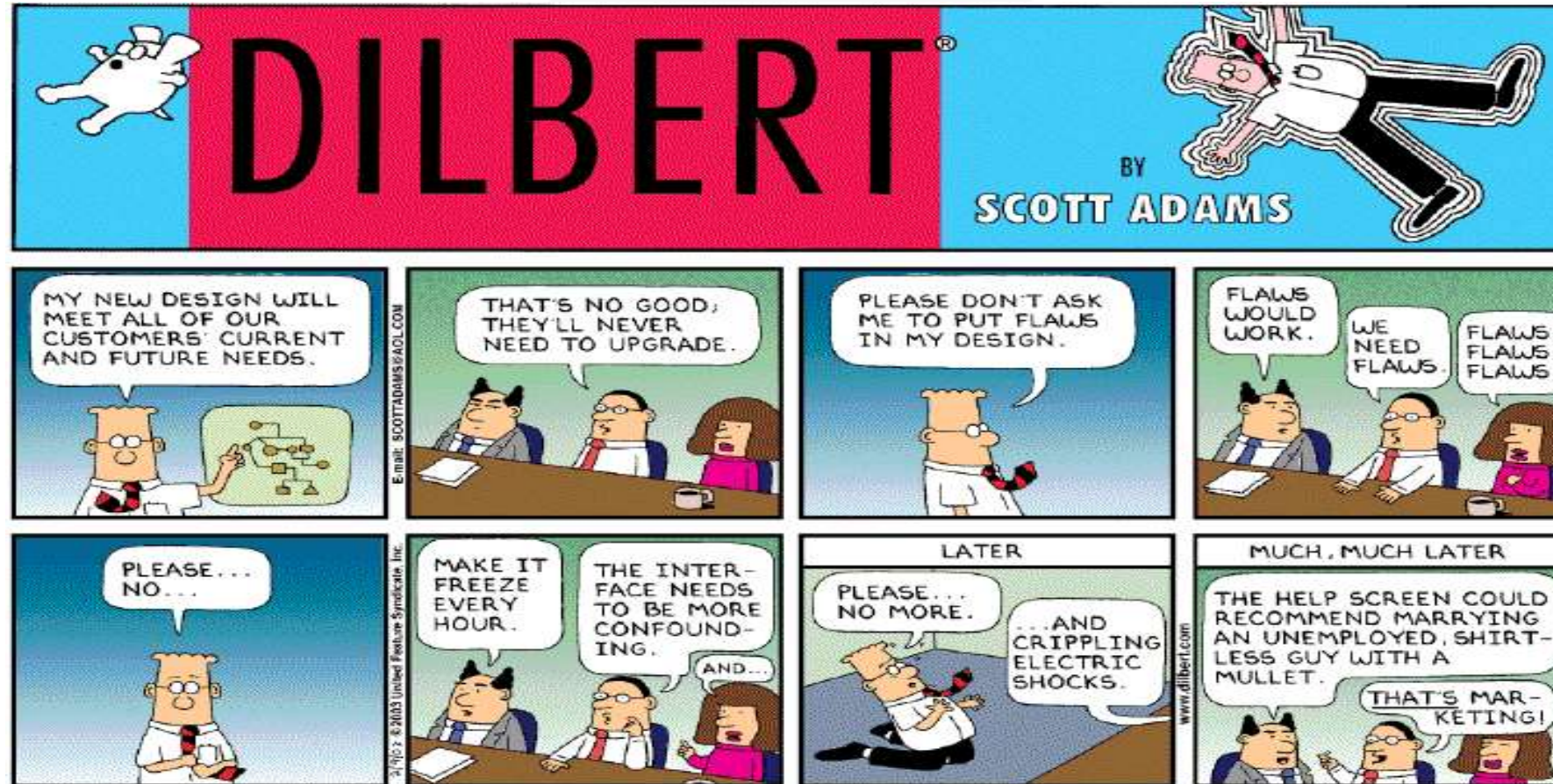
Some practical issues

- Who are the users?
- What are 'needs'?

Target users



Users' needs



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Brief overview of common methods to gather user data

- Interviews
- Questionnaires
- Observation
- Choosing and combining techniques

2. Developing alternative designs

- **Competitive/Comparative Analysis,**

- Try using similar services or products in order to find out:
 - Current trends in the marketplace
 - What expectations your users will have
 - What *to* do, what *not* to do
 - Interface conventions
 - “Must have” standard features

- **Heuristic Evaluation,**

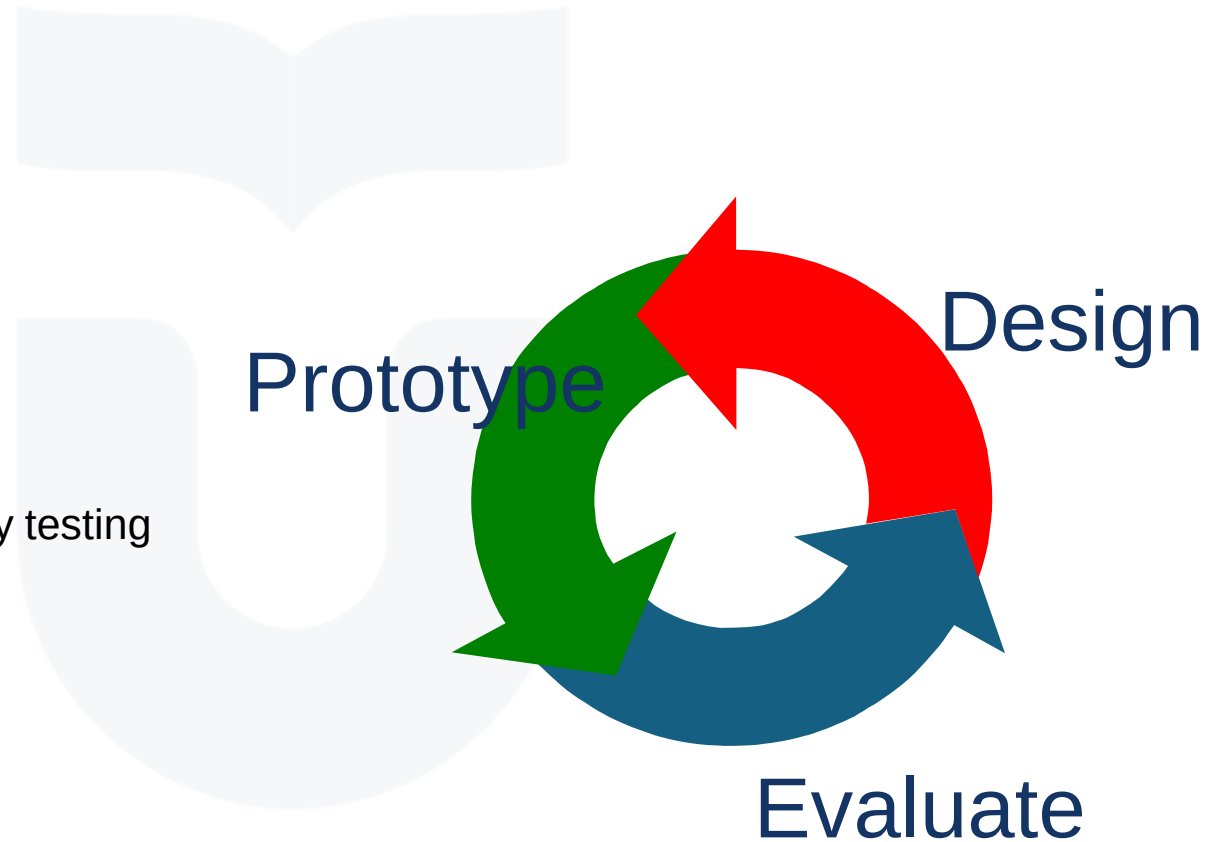
- Evaluate an existing interface (or new interface concept) based on set of usability criteria
- Mostly used to highlight usability problems and deficiencies
- May or may not propose usability solutions
- Identified problem areas are addressed by subsequent design work
- Normally done with expert evaluators, but it can be a valuable tool for anyone
- One detailed checklist: <http://www.stcsig.org/usability/topics/articles/he-checklist.html>

2. Developing alternative designs

- **Persona,**
 - Models of “archetypical” users culled from user research
 - Each persona is a description of one particular “typical” user of your system
 - Personas may be combined if they have the same (or sometimes overlapping) goals
 - Places the focus on specific users rather than on “everyone”
- **Goals, Tasks & Scenarios,**
 - **Goals:**
 - Are what the user wants to do, but not how the user achieves them
 - **Tasks:**
 - Describe the steps necessary to achieve the goals
 - Can vary with the available technology
 - Are broken down into steps for task analysis, and are recombined into sequence of steps for scenario development
 - Designers can reorganize, combine, or remove tasks currently performed to help users achieve their goals more efficiently
 - **Scenarios:**
 - Written description of a persona achieving a goal through a set of tasks in a specific context
 - Should start technology-neutral and become more specific as the design progresses

3. Building interactive versions of the designs

- Start rough
- Explore!
- Use personas to keep the users in view
- Use scenarios to inform the design
- Get frequent feedback
- Note user conventions
- Make design artifacts public
- May be expressed in a prototype for usability testing



4. Evaluating designs

- Let users validate or invalidate the design
- Ask the user to complete selected typical tasks (from scenarios) and think aloud while they do it
- Test early in the process
- Can test with 3-5 users (or less!)
- “Formal” testing
- Measures “success”
 - Set success criteria prior to testing (best done at the project outset)
 - Compare to baseline if you have one
 - Have usability problems revealed in the heuristic evaluation been addressed?

5. Evaluating designs

- Define what is to be tested
- Select users based on personas
- Administer the tests
- Analyze the data
- Document the findings in a brief
- Share the findings with the development team
- Determine what design changes will be made based on test results

Interaksi Manusia dan Komputer (IMK)

THE USER INTERFACE (UI) DESIGN PROCESS

Overview Proses Desain UI



Proses Desain UI (14 Step)

- **Step 1** : Know Users
- **Step 2** : Understand Business Function
- **Step 3** : Understand the Principles of Good Interface and Screen Design
- **Step 4** : Develop System Menu & Navigation
- **Step 5** : Proper Windows
- **Step 6** : Proper Interaction Device
- **Step 7** : Screen-based controls
- **Step 8** : Clear text & message
- **Step 9** : Provide Effective Feedback
- **Step 10** : Internasionalization & Accessibility
- **Step 11** : Create Meaningful Graphics, Icons, and Images
- **Step 12** : Choose Proper Colours
- **Step 13** : Organize & Layout Windows & Page
- **Step 14** : Testing

Step 1 Know User

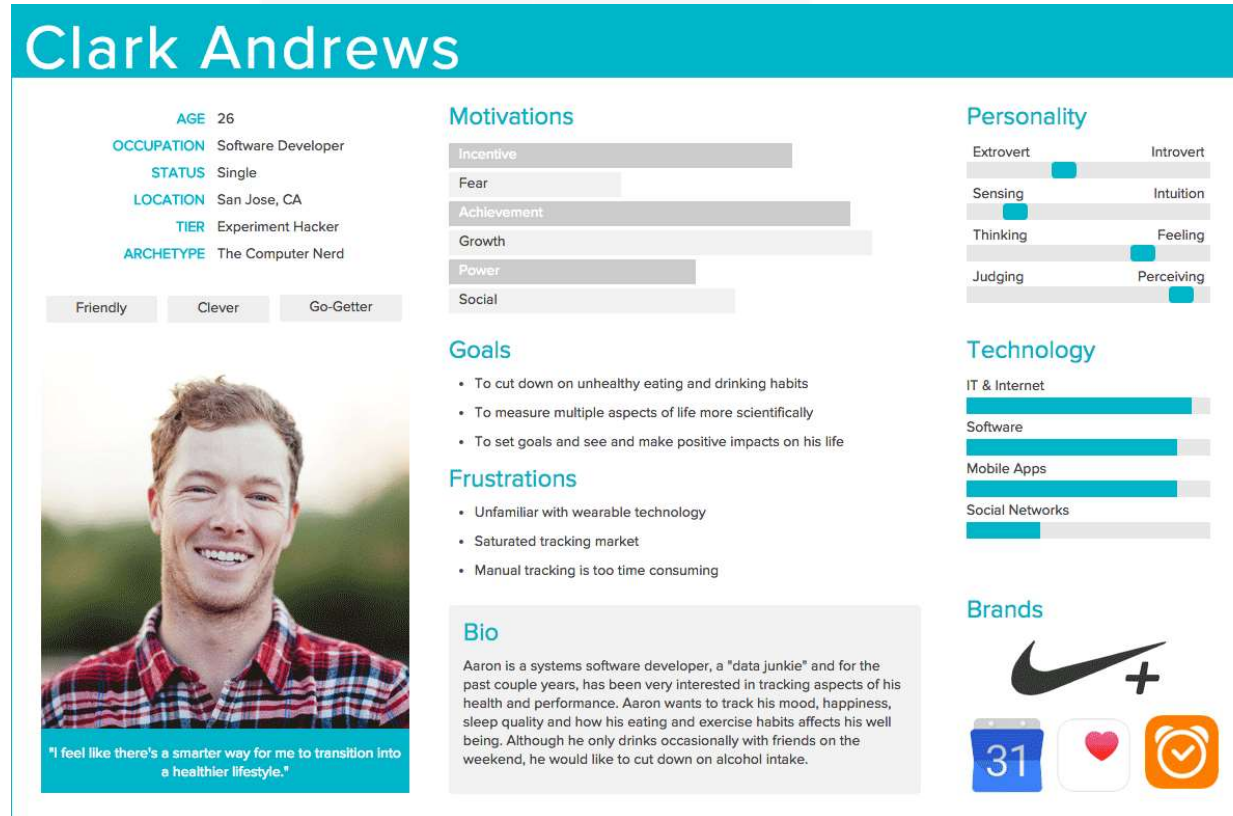
- Tujuan :

1. Memahami pentingnya mengenal pengguna dari aplikasi yang akan dibangun sebagai dasar untuk merancang antarmuka perangkat lunak
2. Mendefinisikan *Persona User* antarmuka perangkat lunak

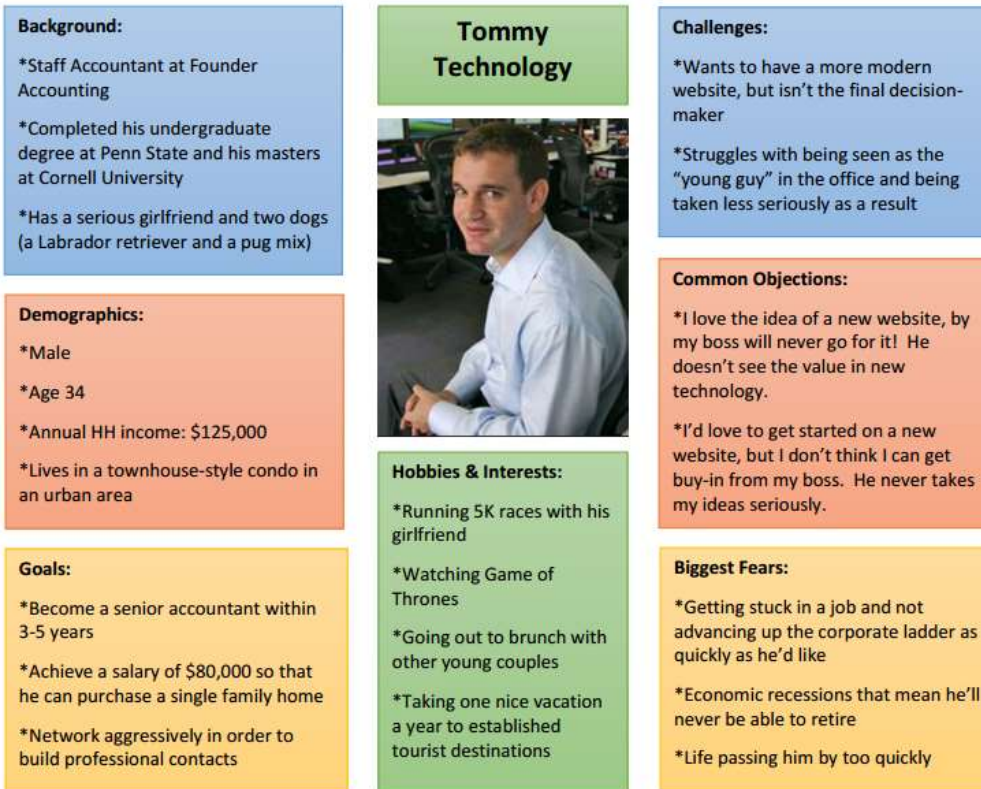
User Persona Example



User Persona Example



User Persona Example



Step 2 Understand Business Function

- Tujuan :

1. Memahami pentingnya mengetahui fungsi bisnis berdasarkan *Persona user* sebagai acuan dalam merancang antarmuka perangkat lunak.
2. Mengerti dan menerapkan setiap tahapan dalam mengidentifikasi alur kerja setiap *task* berdasarkan *Persona User*.

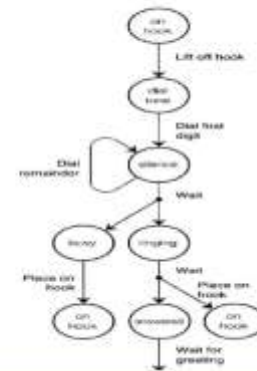


A **mental model** is:
'an explanation of
someone's **thought**
process about how
something works in
the real world'.

Wikipedia

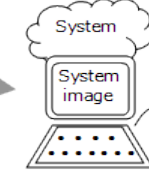


Contoh: Model Mental pada Panggilan Telepon



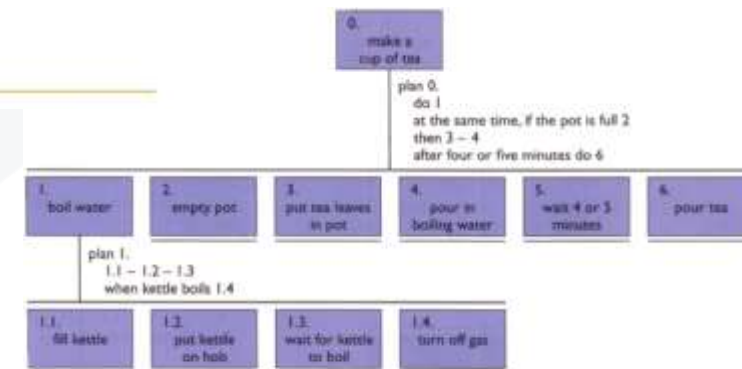
Designer's
mental model

Designer



User's
mental model

User



[Dix et al., p. 515]

Step 3 Understand the Principles of Good Interface and Screen Design

- Tujuan :

Menerapkan prinsip design interaksi dalam merancang antarmuka perangkat lunak untuk menghasilkan antarmuka perangkat lunak yang baik.

- Basic Design Principles :

Consistency, Simplicity, Context

Step 4 Develop System Menu & Navigation

- Tujuan :

Mahasiswa mampu merancang Struktur menu & navigasi yang sesuai dengan prinsip UI



Struktur Menu

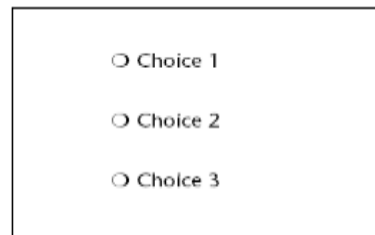
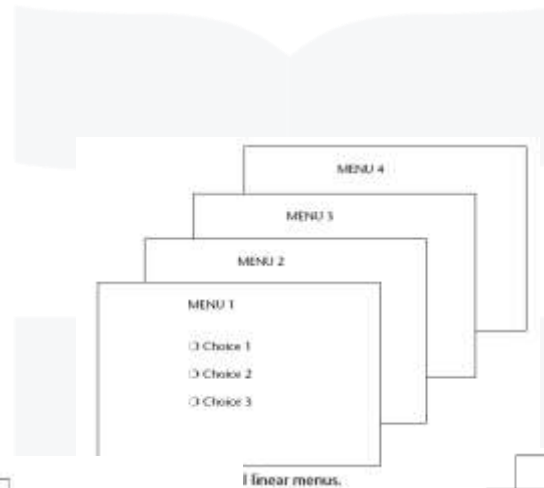


Figure 4.1: Single menu.



Linear menus.

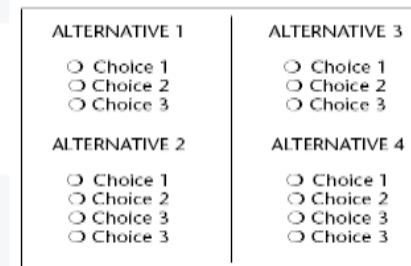


Figure 4.3: Simultaneous menus.

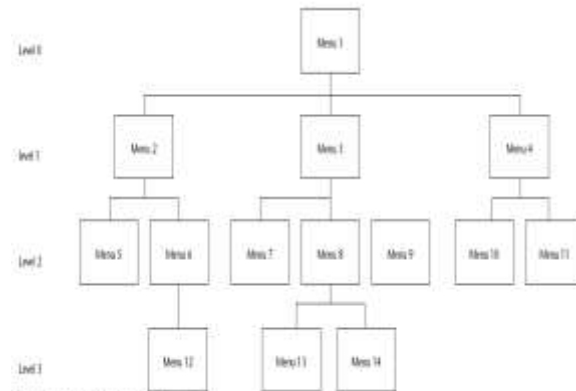


Figure 4.4: Hierarchical or sequential menus.

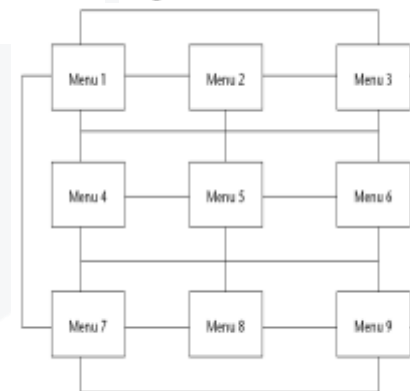
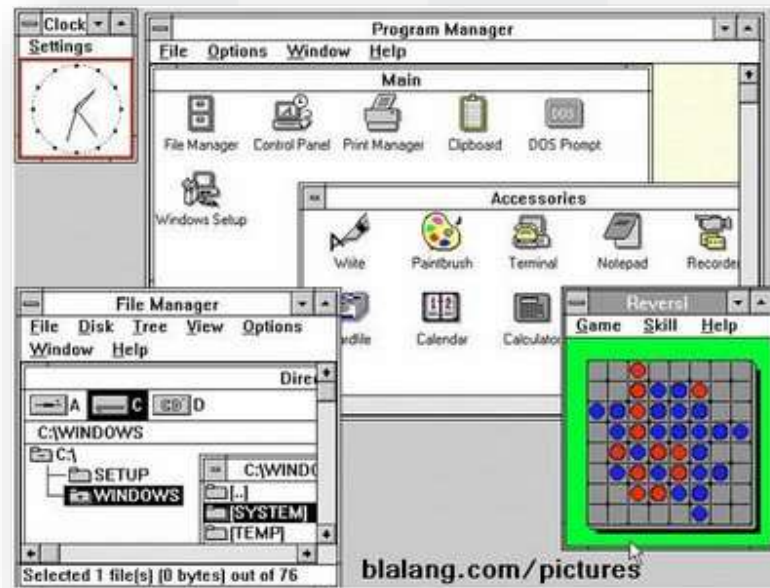


Figure 4.5: Connected menus.

Step 5 Proper Windows

- Tujuan :

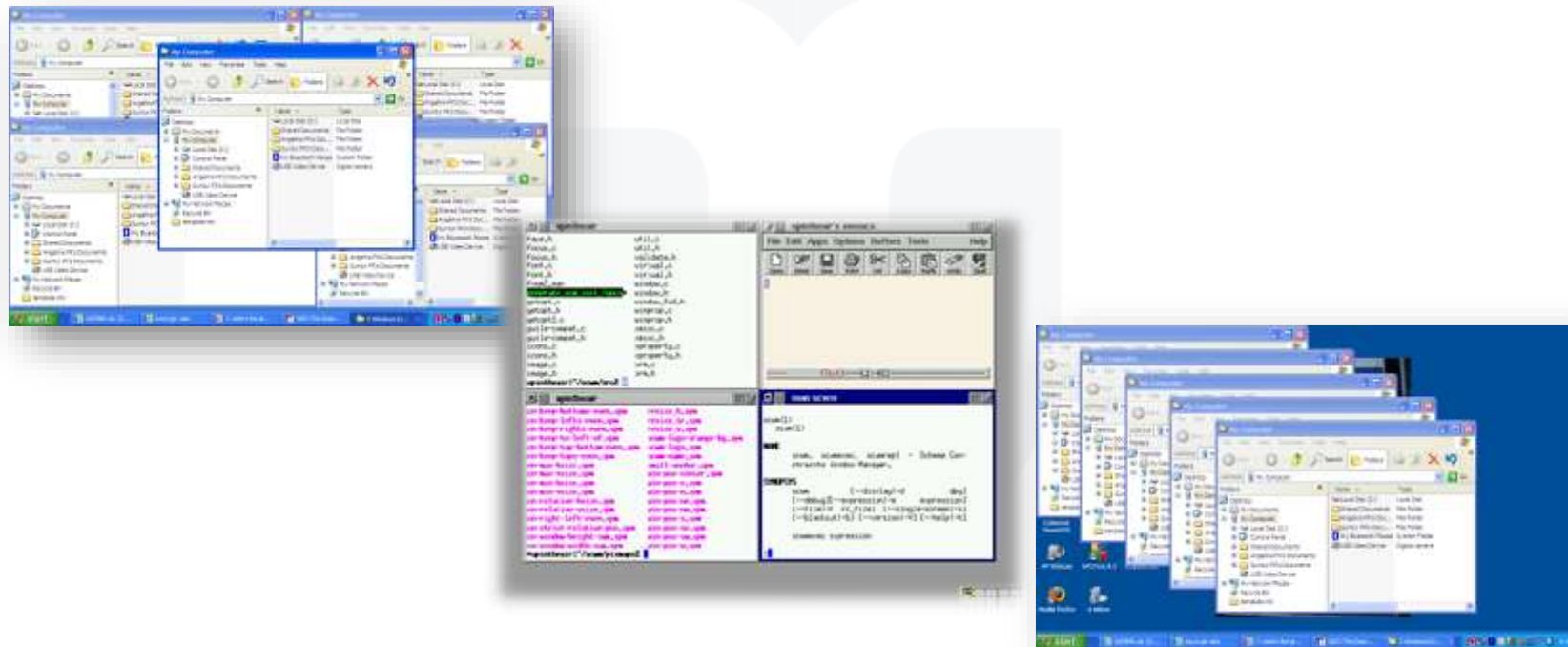
Mahasiswa/i dapat mampu memilih dan merancang windows yang sesuai dengan kebutuhan.



Tipe-tipe Windows

- *Primary windows*
 - Window utama yang muncul di layar saat sebuah aktivitas atau aksi dimulai
 - Disebut juga *application window / main window / parent window*
 - Merupakan titik utama aktivitas pengguna
- *Secondary windows*
 - Windows tambahan yang dapat berupa *independent* maupun *dependent window* terhadap *primary window*-nya
 - Dapat berupa *modal* maupun *modeless*
- *Dialog box*
 - Untuk memperluas dan melengkapi interaksi dengan aksi tertentu yang sangat terbatas
 - Untuk menampilkan pesan, aksi singkat, atau tombol *command* (OK, *cancel*, dll)

Gaya tampilan windows



Step 6 Proper Interaction Device

- Tujuan :

Mahasiswa/i dapat mampu memahami karakteristik setiap perangkat interaksi sehingga dapat memilih perangkat interaksi yang tepat diterapkan dalam desain UI.

SELECTING THE PROPER INPUT DEVICE



Step 7 Screen Based Controls

- Tujuan :

Mahasiswa/i mampu memilih dan menerapkan *screen-based controls* yang tepat dengan kebutuhan

Controls examples

Single occurrence

Composition:

Or

Composition:

Multiple occurrence

Offices:

Read-only / display

Location:

Alice Springs
Kakadu National Park
Traralgon
Wagga Wagga
Whyalla

If alphanumeric then left justify

Balances:

12,642,123.05
53.98
355,125.44
199.13
612.01

If numeric then right justify

Segmentation

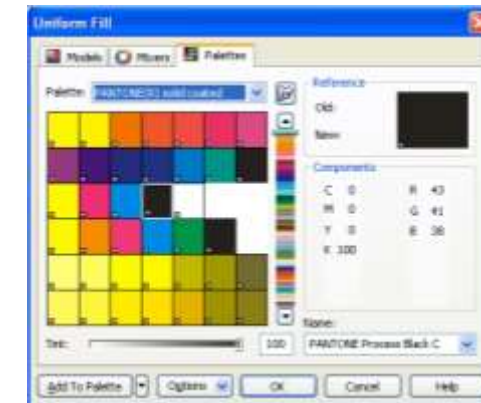
Date:

Telephone:

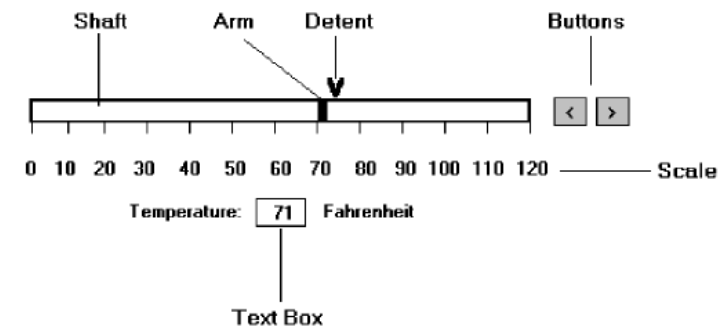
Or

Date: / /

Telephone: () -



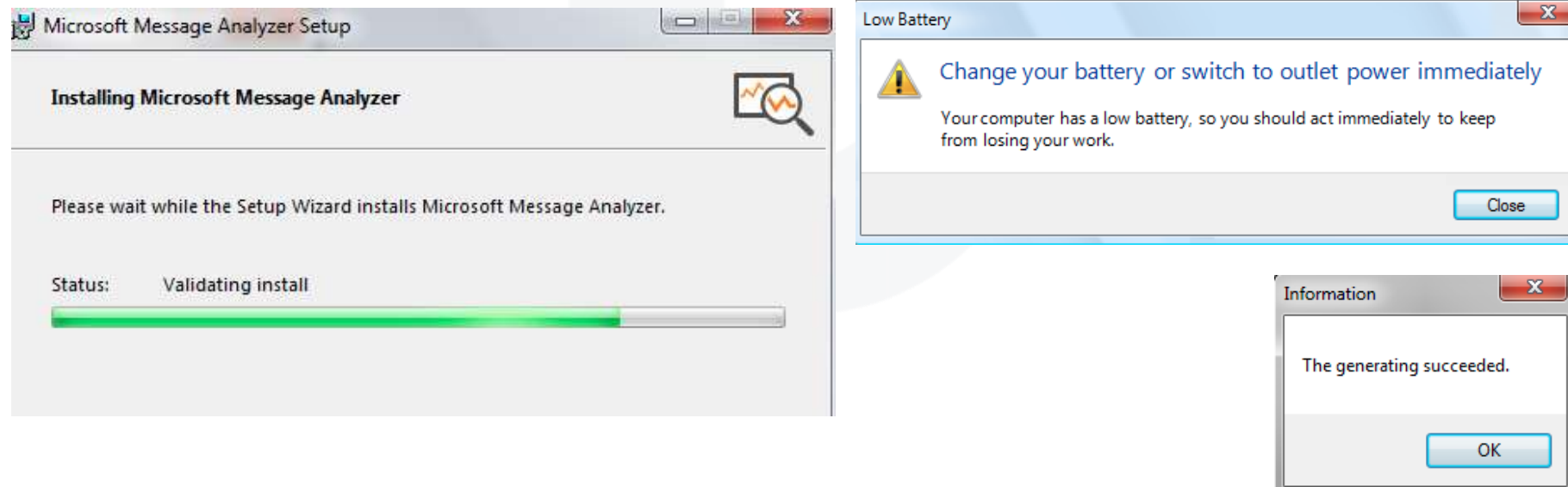
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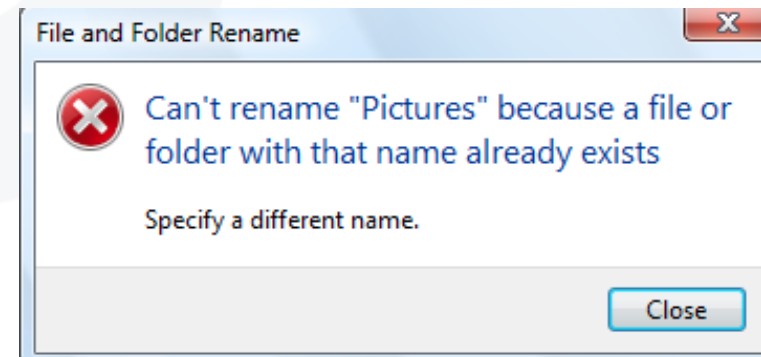
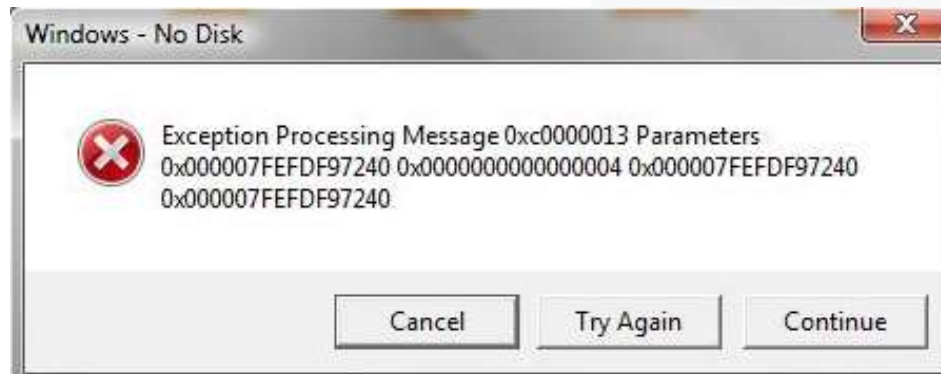
Step 8 Clear Text & Message

- Tujuan :

Mahasiswa/i mampu merancang teks dan pesan yang tepat dan sesuai dengan prinsip UI



Which you prefer? Why?

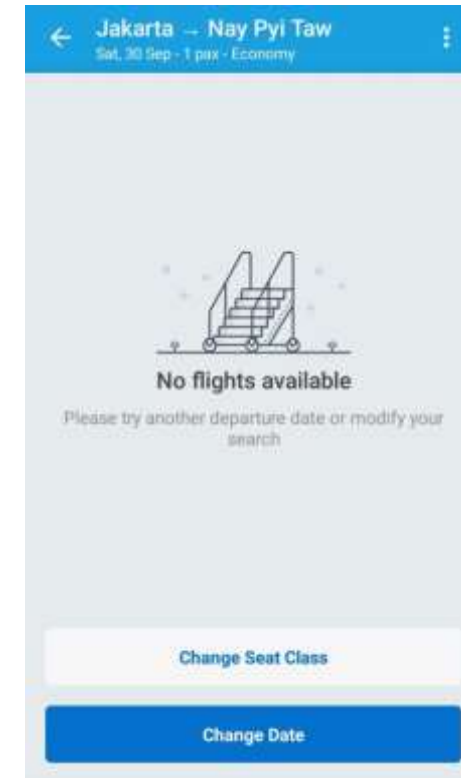


Step 9 : Provide Effective Feedback

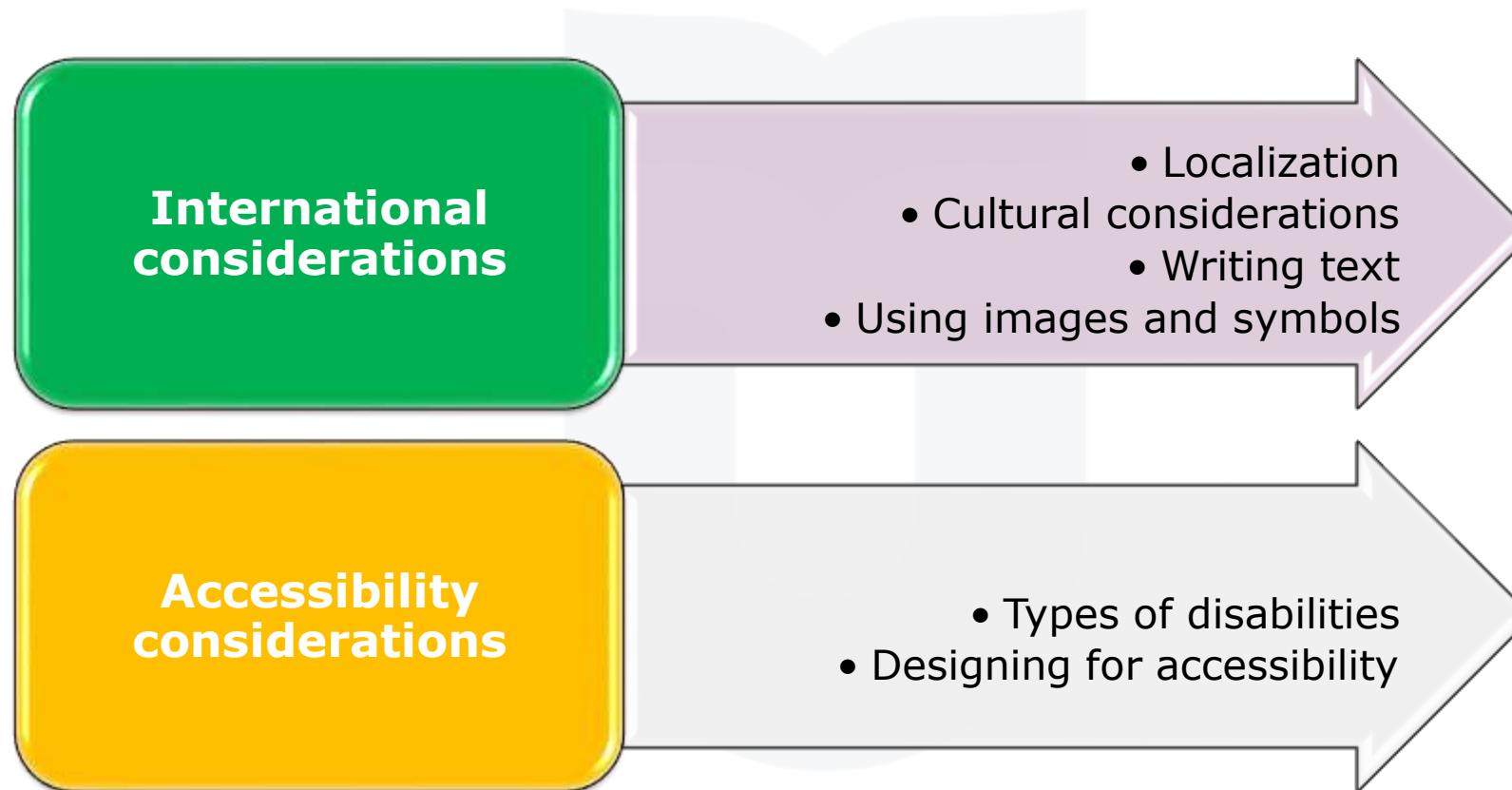
- Tujuan :

Mampu merancang feedback, guidance, dan assistance dengan efektif .





Step 10 Internasionalization & Accessibility



Graphics

- **Images:** should convey their intended messages
- **Image maps:** to provide navigation links to other content
- **Photographs/ Pictures:** used when every aspect of the image is relevant
- **Video:** to show things moving/ changing over time
- **Diagrams:** to show the structure/ relationship of objects
- **Drawings:** when selected parts need to be emphasized/ represented
- **Animation:** to explain ideas involving a change in time/ position
- **Audition:** as a supplement to text and graphics

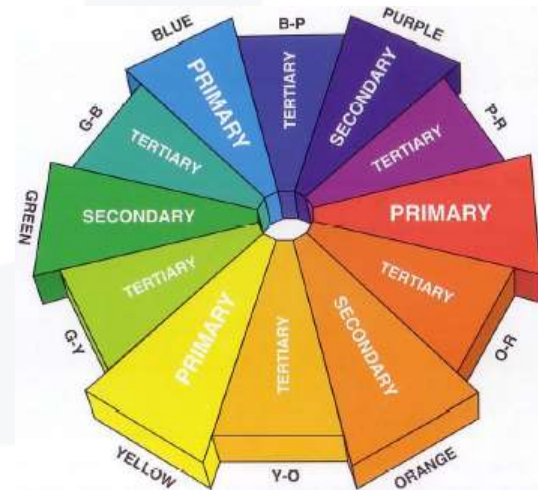
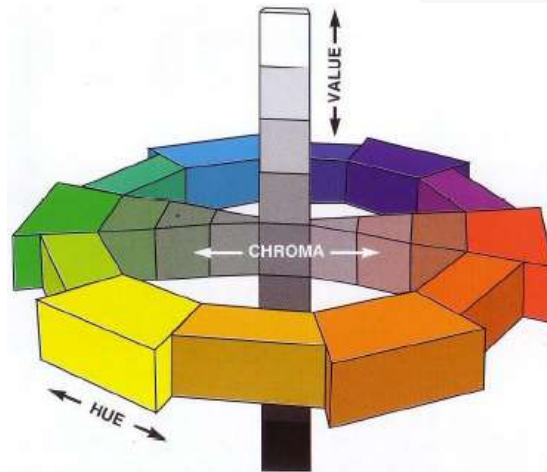
MEDIUM	PERCENT MORE LEARNING
Hearing spoken text and viewing graphics	91%
Viewing graphics alone	63%
Viewing text and viewing graphics	56%
Hearing spoken text, viewing text, and viewing graphics	46%
Hearing spoken text and viewing text	32%
Viewing text alone	12%
Hearing spoken text alone	7%

From Lee and Bowers (1997).

Step 12 Choose Proper Colors

- Tujuan :

Mahasiswa/I mampu menerapkan warna yang sesuai dengan kebutuhan



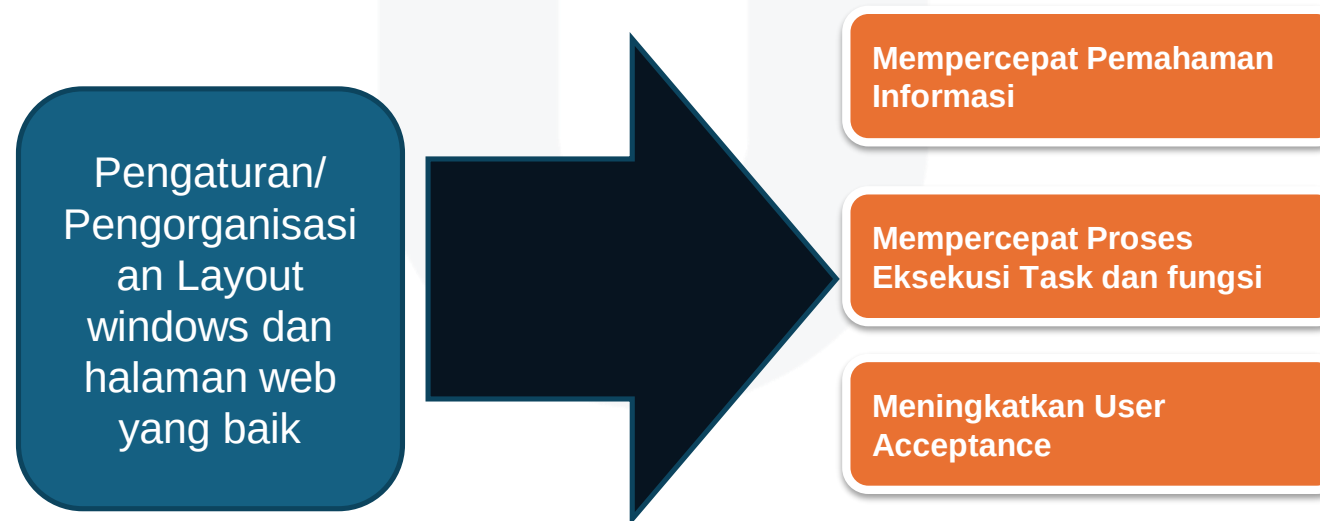
Color

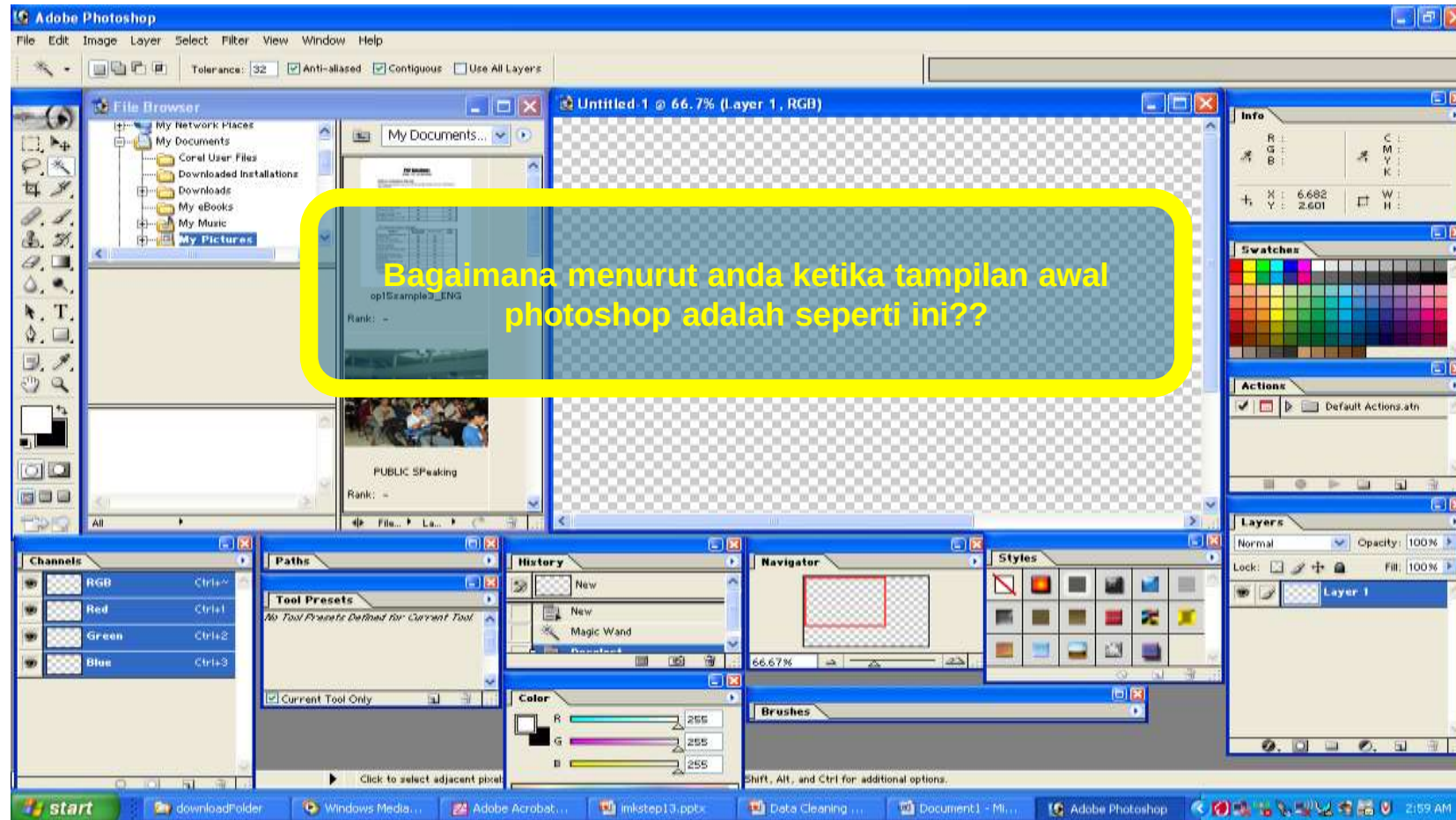
- **Color** adds dimension, or realism, to screen usability.
- Color draws **attention** because it attracts a person's eye.
- If used **properly**, it can emphasize the logical organization of information, facilitate the discrimination of screen components, accentuate differences among elements, and make displays more interesting and attractive.
- If used **improperly**, color can be distracting and possibly visually fatiguing, impairing the system's usability.

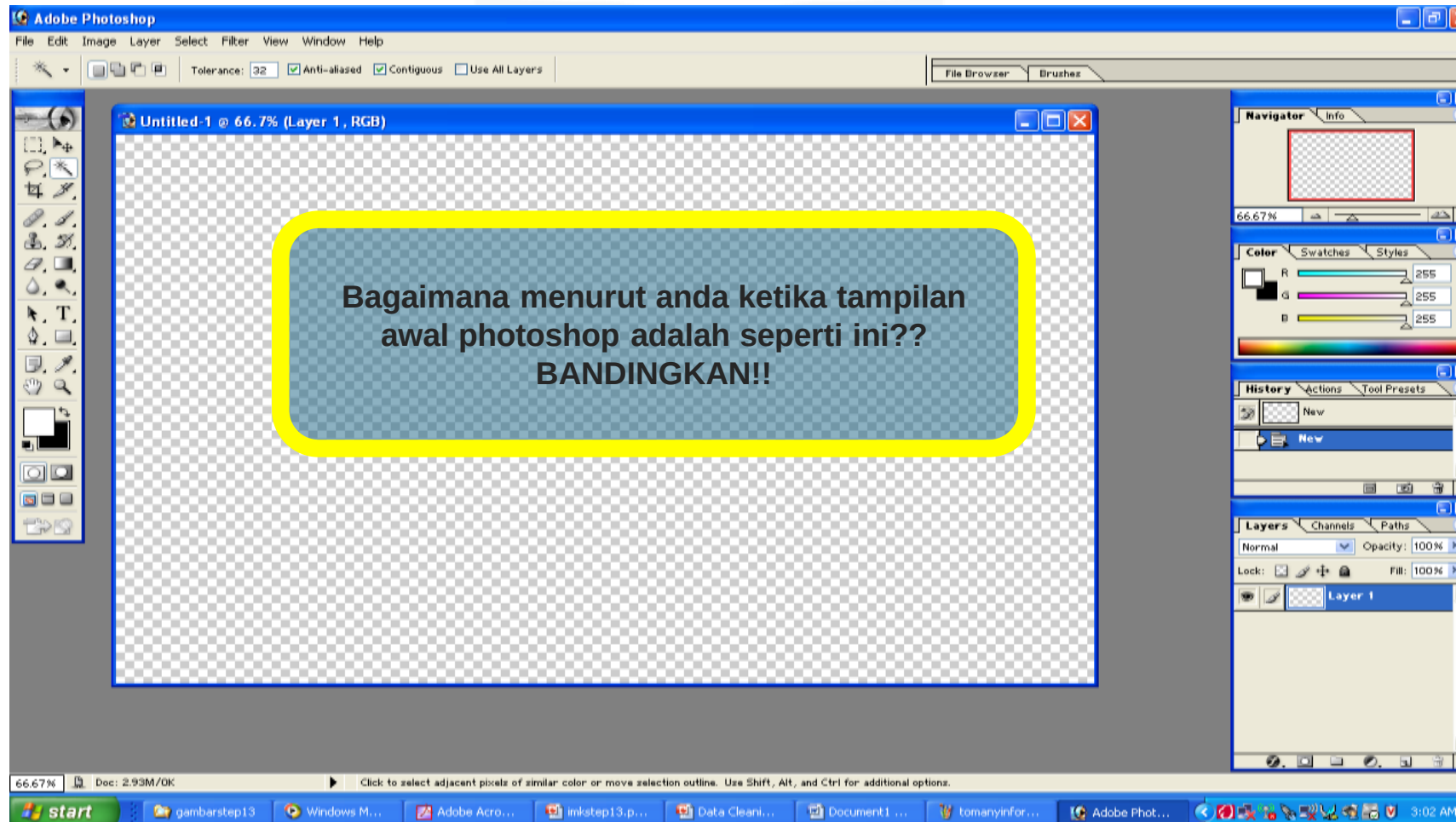
Step 13 Organize & Layout Windows & Page

- Tujuan :

Mahasiswa/i dapat mampu menerapkan pengaturan dan tataletak Windows dan halaman web yang sesuai dan bermakna.







Step 14 Testing

Tujuan :
Mahasiswa/i dapat memahami konsep pengujian
dalam desain UI dan cara mengukur UI



Bahan bacaan

- Wilbert O. Galitz, 2007, **The Essential Guide to User Interface Design: An Introduction to GUI Design Principles and Techniques, Third Edition**, Wiley Publishing, p.71-102 (dan untuk pertemuan minggu depan p.103-126)
- <https://uxmastery.com/what-does-a-user-centred-design-process-look-like/>
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- <http://www.smashingmagazine.com/2014/08/a-closer-look-at-personas-part-1/>
- <http://webdesign.tutsplus.com/articles/defining-and-applying-personas-to-ux-design--webdesign-7561>
- <http://www.storyboardthat.com/articles/software-development/using-personas-to-identify-your-target-customers>